

Kolej Matrikulasi Negeri Sembilan
DC014 COMPUTER SCIENCE
PRE-PSPM 2024/2025
(2 HOURS | 70 MARKS)

Name : _____

Practicum : _____

1. (a) Multimedia technology has a significant impact on communication, particularly through the widespread use of multimedia elements beyond text on social media platforms.

Describe **one (1)** other multimedia elements that can enhance communication on social media?

[2 marks]

- (b) State the purpose of the multimedia elements used in the following scenario.

[2 marks]

Scenario	Purpose of Multimedia Element
Les' Copaque Production brings the characters of the talking mouse deer, crocodile, and tiger to life in the story "Pada Zaman Dahulu."	
Police investigating officers try to draw the face of a criminal suspect based on the victim's testimony using an iPad	

- (c) Based on **FIGURE 1.1**, differentiate between image **J** and **S** in terms of their axis of multimedia application representation.

[2 marks]



J



S

FIGURE 1.1

J: _____

S: _____

2. A successful multimedia project is developed through three phases: Pre-production, Production, and Post-Production.

(a) "The team reviews the finished product thoroughly, checking for errors, ensuring smooth transitions, and gathering feedback to confirm that all elements meet quality standards before release."

- (i) Identify the phase of multimedia design from the given scenario.

[1 mark]

- (ii) Describe your answer in **2(a)(i)**

[2 marks]

- (iii) Based on your answer in **2(a)(i)**, give **one (1)** activity involved in this phase.

[1 mark]

- (b) "Authoring tools are software applications used to create digital content, like e-learning courses, presentations, and multimedia projects."

- (i) Differentiate the types of authoring tools according to the criteria listed in the table below.

[6 marks]

Authoring Tools	Characteristic	Example Software
Icon-based		
Page/Card-based		

- (ii) Explain the type of navigation as shown in **FIGURE 2.1**

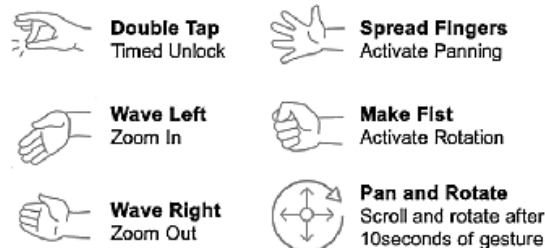


FIGURE 2.1

[2 marks]

- (c) Identify the suitable medium of delivery for the situation described below.

[4 marks]

Description	Medium of Delivery	Example
A teacher uploads lessons and quizzes to an e-learning platform for students to access from home.		
A trainer gives participants a USB drive with all course materials to use.		

3. The One Selamat company wants to move its current file-based system to a database system. In many ways, this can be seen as a good decision.

- (a) Give **two (2)** differences between database systems and file processing systems.

[4 marks]

Database System	File Processing System

- (b) Explain **one (1)** advantage if the company converts to a database system.

[2 marks]

4. "Data integrity is the process of ensuring an organization's data is accurate, complete, consistent, and reliable."

- (a) Explain **one (1)** integrity risks that the organization could get if their database doesn't have data integrity.

[2 marks]

- (b) How the organization can minimize integrity risks based on your answer in 4(a).

[2 marks]

5. A network is a collection of interconnected devices or nodes that communicate with each other to share resources, exchange data, and facilitate communication.

- (a) Network media refers to the physical or logical means through which data is transmitted in a network.

- (i) Explain two types of transmission media based on the table below.

[4 marks]

Criteria	Explanation
Guided media	
Unguided media	

- (ii) Identify which type of transmission media would be more suitable for a temporary event, such as a conference or trade show. Justify your answer.

[3 marks]

- (iii) Give two examples of the type of transmission media that you mention in **5 (a)(ii)**.

[2 marks]

- (b) An educational institution seeks a high-speed wired network upgrade to connect multiple buildings and support online classes for faculty and students.

- (i) Identify suitable transmission media the institution should choose for this upgrade.

[1 mark]

- (ii) Describe transmission media mention in **5 (b) (i)**

[2 marks]

- (c) **FIGURE 5.1** below shows a basic network diagram.

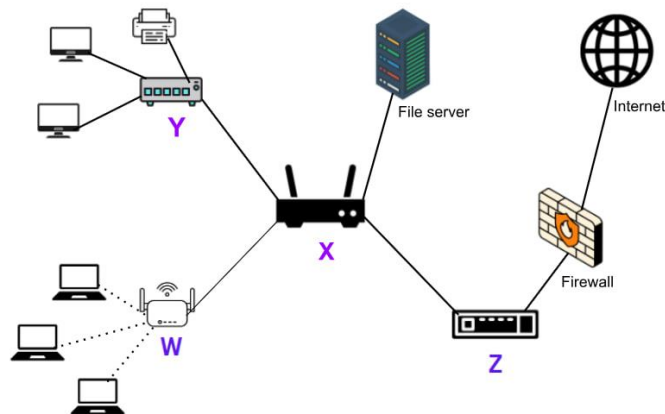


FIGURE 5.1

- (i) Z is a main device used to connect to the Internet. Y is a networking device that connects multiple devices within a local area network and X is a device that connects multiple devices to a network and directs data traffic between them.

Identify X, Y, and Z.

[3 marks]

X : _____

Y : _____

Z : _____

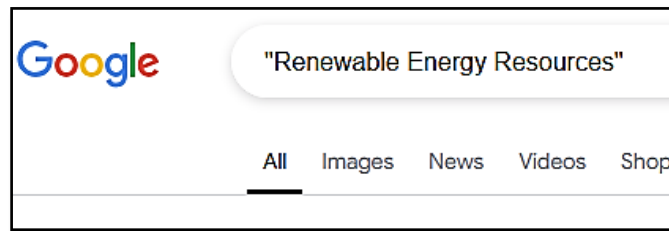
- (ii) Describe W.

[2 marks]

- (iii) Identify **one (1)** communication device that enables a desktop computer to connect to a network.

[1 mark]

6.

**FIGURE 6.1**

- (a) **FIGURE 6.1** shows that Amin Fattah is doing a project on "Renewable Energy Resources" for his class. He needs to find reliable sources, images, and recent news articles to support his research.

- (i) Which is the best search tool Amin can use to access a wide range of reliable information online?

[1 mark]

- (ii) Describe your answer in **6 (a)(i)**.

[2 marks]

- (b) Developing effective search strategies is essential for students to quickly find accurate information and excel in academic research.

- (i) Choose the suitable search strategies that use quotation marks.

[1 mark]

- (ii) Explain why your answer in **6(b)(i)** can help student find information more effectively and efficiently.

[2 marks]

- (iii) Give **one (1)** example of the search strategy.

[1 mark]

- (c) Identify the suitable search strategies used in the following scenario:

[2 marks]

Zara is researching how social media affects mental health and is looking for articles on "depression" or "anxiety."	
Kumar is searching for recent reports on climate change policies in Negeri Sembilan from government websites.	

- (d) **FIGURE 6.2** presents the statistics of popular social media platforms in Malaysia, as reported by the Shopify website.

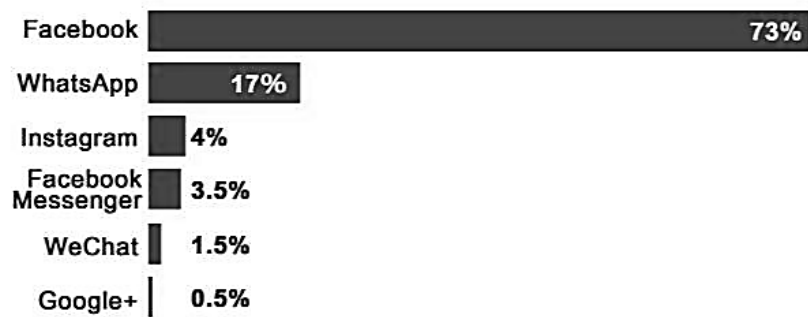


FIGURE 6.2

- (i) Give a brief description of social media [2 marks]

- (ii) State **one (1)** other web activity that you use daily other than social media. [1 mark]

- (iii) List **one (1)** example of applications from your answer in 6(d)(ii) [1 mark]

(e)

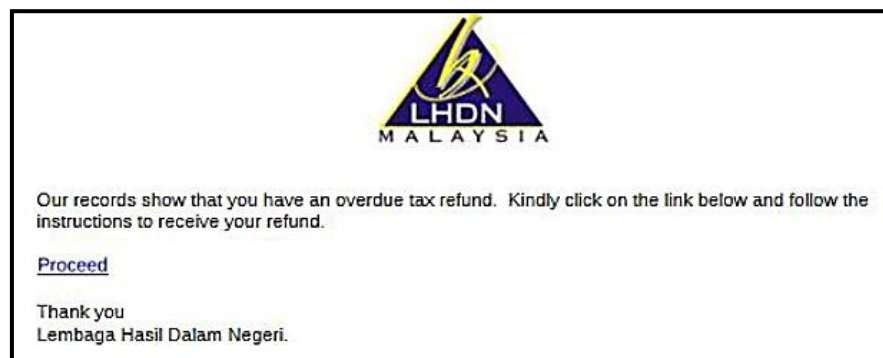


FIGURE 6.3

FIGURE 6.3 shows that phishing remains one of the most frequent activities, often targeting users through deceptive emails in an attempt to obtain sensitive information.

- (i) Describe one way to implement precautions against phishing in emails. [2 marks]

- (ii) State one importance of safeguarding our personal information. [2 marks]

- (f) The Internet of Things (IoT) connects everyday devices to the internet, allowing them to gather and share data, making technology more efficient and improving our daily lives.

- (i) Explain one way the Internet of Things (IoT) is used in daily life? [2 marks]

- (ii) Artificial Intelligence (AI) enhances the functionality of the Internet of Things (IoT) by enabling connected devices to analyse data and make intelligent decisions automatically.

State **one (1)** example of how AI is used in everyday life to make things easier or more efficient?

[1 mark]

END OF QUESTIONS